S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

# RiskPulse

A real-time AI/NLP risk engine that reads news and social posts, turns each into a structured risk signal, and acts on it: a tactical index rebalancer and a wholesale-bank stress test.



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# Risk desks see the news; they cannot score all of it

## The problem

Thousands of headlines and posts a day, in free text

Analysts triage by hand; stress tests run on fixed schedules

A static shock ignores what an event actually does to rates, spreads and sectors

1

### Read

News (GDELT, RSS) and social posts, deduplicated into stories

2

### Score

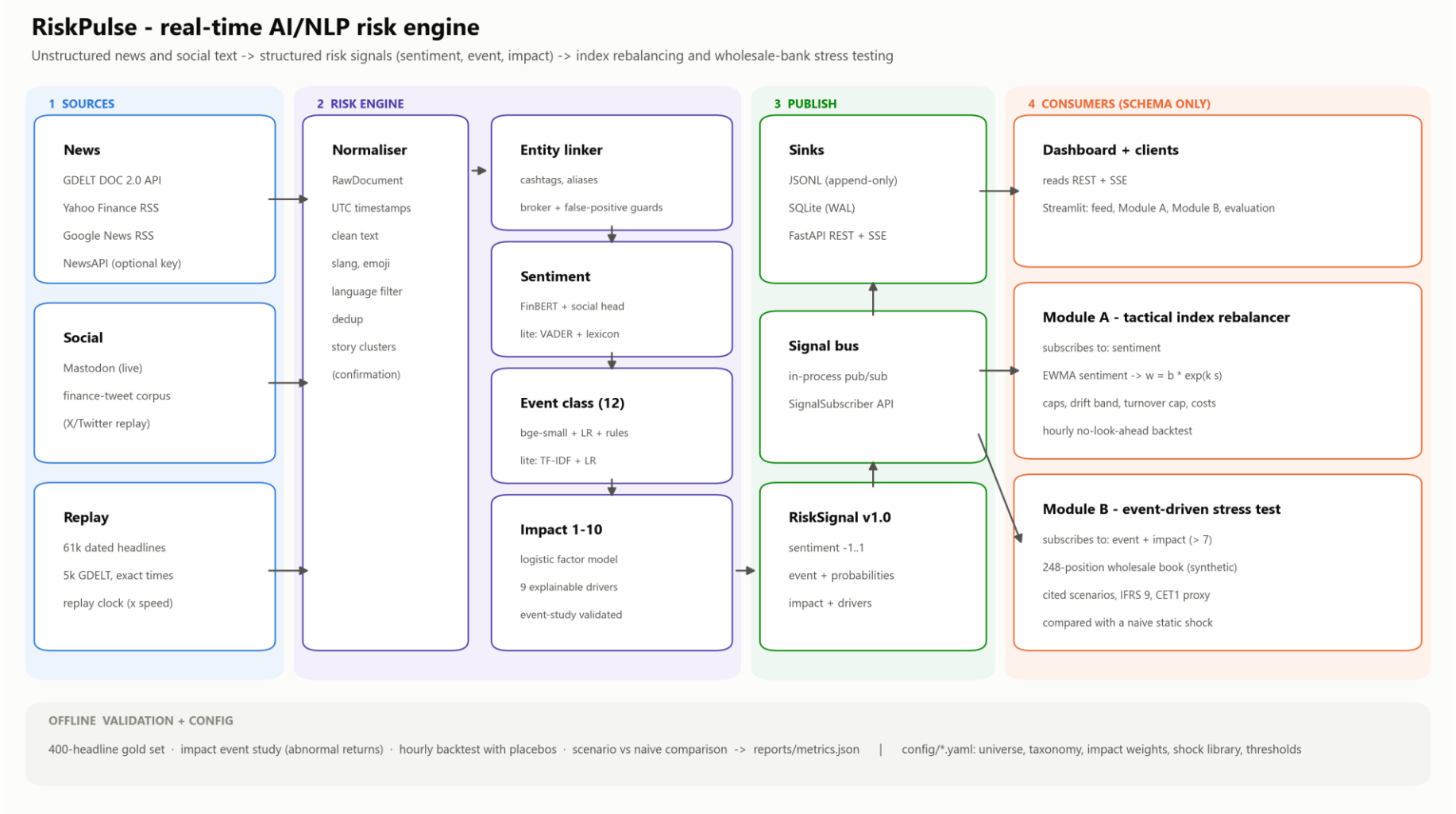
Sentiment −1…1, one of 12 risk events, impact 1–10 with its drivers

3

### Act

Module A tilts index weights; Module B stress-tests a $8.5bn book

# One engine, decoupled consumers, one versioned signal



# Built for a risk desk: explainable, testable, offline-first

### Hybrid NLP

FinBERT + a tweet-trained head blended by source; 12-class event model with high-precision rules

### Explainable impact

Logistic factor model: 9 drivers stored on every signal, validated by an event study

### Decoupled design

Modules read only the RiskSignal schema via in-process, SQLite or SSE subscribers

### Module A

Sentiment tilt with caps, drift band, turnover cap and costs; hourly no-look-ahead backtest

### Module B

Fires only on reported adverse events; cited scenarios scaled by impact, IFRS 9 staging, Basel CRE20 weights, CET1 proxy

### Runs anywhere

Lite tier without torch, offline replay demo, 114 tests, CI on Python 3.11

# Measured against naive baselines on held-out data

**0.69** vs 0.46

sentiment macro-F1, 200 held-out headlines (naive: VADER)

**0.65** vs 0.51

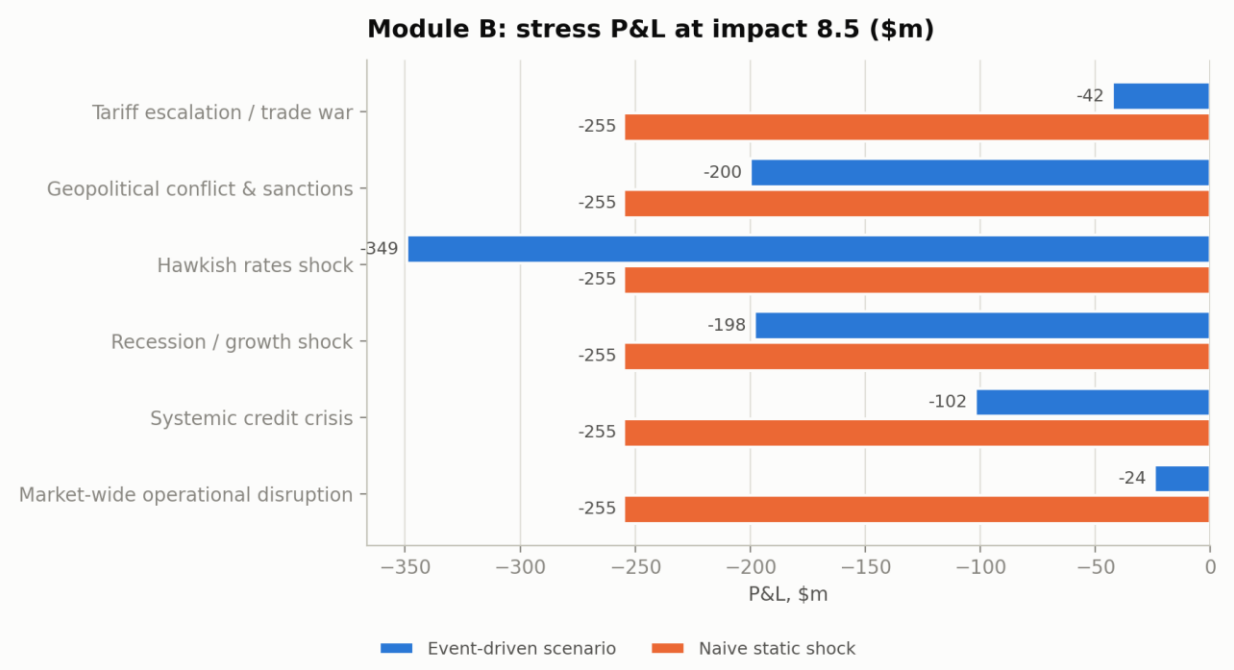
event macro-F1 on the same headlines (naive: keyword rules)

**1.21×** vs 0.85×

S&P 500 move on red-alert days, out of sample (naive: busiest news days)

**4 / 6** scenarios

where a naive static stress test gets the rates P&L sign wrong




**Headline to stress result:** 0.2 s on a laptop CPU, vs an overnight batch run (assumed).

**Triage:** ~84 headlines a day become 1.5 stress tests; the trigger rules cut 70% of high-impact alerts.

**Honest result:** with strict timing, sentiment tilts show no next-day edge (IC -0.004); an early IR of 3.7 was a look-ahead leak we traced and fixed.

# From headline to capital impact in seconds

### Risk managers

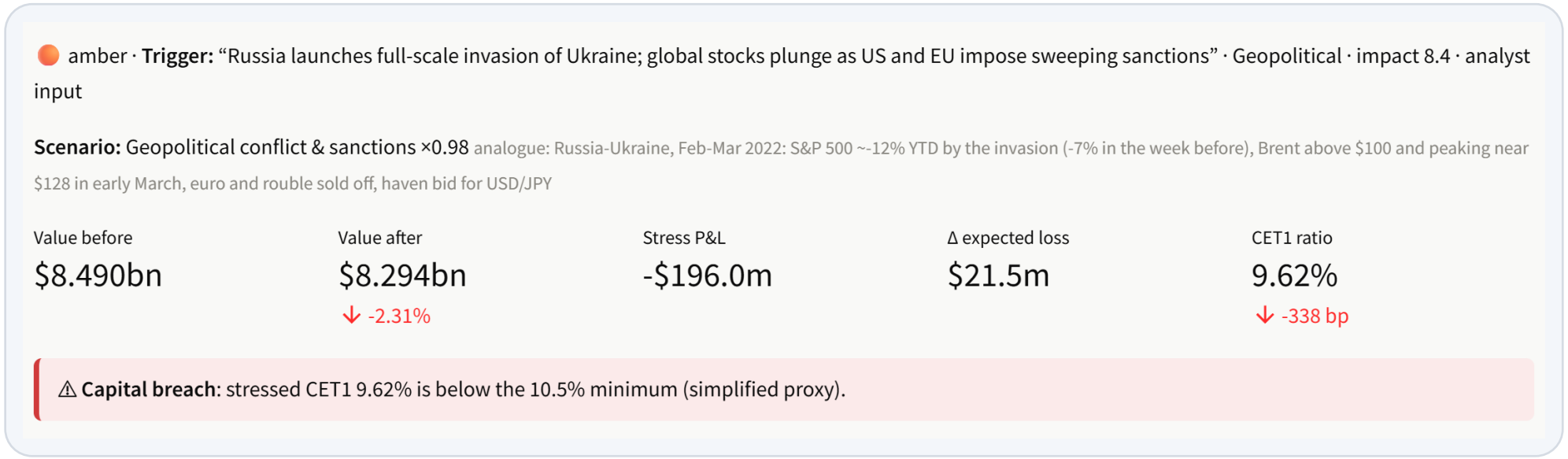
The right stress test runs when an adverse event is reported (not on commentary), with CET1 headroom and the positions that drive the loss

### Portfolio managers

Index tilts react to company news within limits, every change linked to its headline

### Model risk & audit

Every score carries its drivers, schema and model version; results are reproducible from the repo




*Module B in the dashboard: the triggering headline, the scenario applied, value before and after, and the capital check*

# What was hard, what is missing, what comes next

## Challenges we solved

A backtest information ratio of 3.7 was a look-ahead leak: traced and fixed

97% of 61,399 headlines carry a date, not a time: used only after the day ends

Questions and opinion pieces fired stress tests: now filtered (29 skipped in the replay)

The stress trigger ran on 90% of trading days: added a red-alert tier, tested out of sample

## Limitations

Day-level history, so fast intraday reactions are untested

16 mega-cap names and two years of data

Gold labels from one annotator (AI-proposed, author-reviewed)

Sensitivity-based revaluation; simplified CET1 proxy

## Next steps

Intraday timestamped feeds to test fast reactions

Multilingual sources and target-aware sentiment

Kafka bus, feature store, model registry

Full revaluation library and regulatory scenarios



Open source (MIT) · public and synthetic data only · github repo and demo video in the README